

FAST MODEL PREDICTIVE CONTROL OF MINIATURE HELICOPTERS

FINAL PROJECT FOR THE COURSE - EXTENSION

“MODELING AND CONTROL OF CYBER-PHYSICAL SYSTEMS II” – UNITS

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This paper presents an extension to the previous work on Model Predictive Control (MPC) for miniature helicopters. The core contribution of this extension is a structural redesign in the handling of the reference trajectory, transitioning from continuous sampling to a dynamically feasible piecewise-constant (PWC) formulation within the prediction horizon. Moreover, the simulation framework is expanded by introducing new flight shapes such as lemniscate and spiral ellipse. Finally, different initial conditions and sampling times are tested to evaluate the controller’s robustness, accuracy, and convergence.

Introduction

SECTION 1

Introduction

This work presents an extension of the original **Fast Model Predictive Control of Miniature Helicopters**, with the objective of further analyzing its flexibility, robustness, and behavior under more challenging conditions. The first contribution of this extension is the generation of two additional complex flight shapes, the *Lemniscate* and the *Spiral Ellipse*, which enhance the set of reference trajectories and introduce more challenging geometric and dynamic characteristics. These trajectories are defined analytically through differential flatness, ensuring the generation of dynamically feasible state and input references.

Another objective of this study is to assess the effect of representing the reference control inputs in a Piece-Wise Constant (PWC) form instead of a continuous-time formulation. To this end, an **open-loop comparison** is performed between trajectories generated directly from the flatness map, where state and input are sampled continuously, and trajectories obtained by initializing the system at the starting point and then propagating it forward using inputs sampled from the continuous trajectory at discrete time steps.

In order to consistently track these PWC references, the **MPC controller** is modified so that it directly receives pre-generated state and input trajectories, while preserving the original linear time-varying formulation and optimization structure. The extended framework is then evaluated through a comprehensive tracking comparison, considering multiple trajectory shapes, different initial conditions, and various sampling times.

Finally, a **convergence study** is performed, analyzing the evolution of the position error during the simulations. This analysis provides a systematic assessment of the controller's ability to drive the system toward the desired trajectory under different operating configurations.

System Modeling and Planning

PART

II

SECTION 2

Trajectory Generation

Building upon the previous research, this study introduces new complex flight trajectories. This section outlines their definitions and presents their corresponding plots.

SUBSECTION 2.1

Lemniscate Trajectory

The lemniscate trajectory describes a smooth “Figure-8” motion in the horizontal plane at a constant altitude. It is particularly useful for evaluating the controller’s tracking performance and robustness during continuous, alternating coordinated turns. The trajectory is defined in `ShapeLemniscate.m`.

To simplify the notation, let $A = R\sqrt{2}$ and the denominator $D(t) = \sin^2(\omega t) + 1$. The flat outputs $\mathbf{z}(t)$ are defined as:

$$\begin{cases} z_1(t) = \frac{A \cos(\omega t)}{D(t)} + c_x \\ z_2(t) = \frac{A \cos(\omega t) \sin(\omega t)}{D(t)} + c_y \\ z_3(t) = h \\ z_4(t) = \text{atan2}(\dot{z}_2(t), \dot{z}_1(t)) \end{cases} \quad (2.1)$$

where (c_x, c_y) represents the center of the figure, h is the constant altitude, and the yaw angle $z_4(t)$ is constrained to be tangent to the trajectory path.

The first derivatives (velocities) are calculated as:

$$\begin{cases} \dot{z}_1(t) = \frac{A\omega (\sin^3(\omega t) - 3 \sin(\omega t))}{D(t)^2} \\ \dot{z}_2(t) = \frac{A\omega (1 - 3 \sin^2(\omega t))}{D(t)^2} \\ \dot{z}_3(t) = 0 \\ \dot{z}_4(t) = \frac{\dot{z}_1(t)\ddot{z}_2(t) - \dot{z}_2(t)\ddot{z}_1(t)}{\dot{z}_1(t)^2 + \dot{z}_2(t)^2} \end{cases} \quad (2.2)$$

The second derivatives (accelerations) are defined as:

$$\begin{cases} \ddot{z}_1(t) = \frac{A\omega^2 \cos(\omega t) (-3 + 12 \sin^2(\omega t) - \sin^4(\omega t))}{D(t)^3} \\ \ddot{z}_2(t) = \frac{2A\omega^2 \sin(\omega t) \cos(\omega t) (3 \sin^2(\omega t) - 5)}{D(t)^3} \\ \ddot{z}_3(t) = 0 \\ \ddot{z}_4(t) = \frac{(\dot{z}_1 \ddot{z}_2 - \dot{z}_2 \ddot{z}_1)(\dot{z}_1^2 + \dot{z}_2^2) - 2(\dot{z}_1 \ddot{z}_2 - \dot{z}_2 \ddot{z}_1)(\dot{z}_1 \ddot{z}_1 + \dot{z}_2 \ddot{z}_2)}{(\dot{z}_1^2 + \dot{z}_2^2)^2} \end{cases} \quad (2.3)$$

The default configuration parameters for the simulated scenarios are:

- radius scale parameter $R = 1.0$ m;
- angular velocity $\omega = 0.7$ rad/s;
- altitude $h = 1.0$ m;
- center coordinates $(c_x, c_y) = (0, 0)$;
- trajectory duration $T_{\text{lemniscate}} = \frac{2\pi}{\omega}$ s, which corresponds to the completion of one lap.

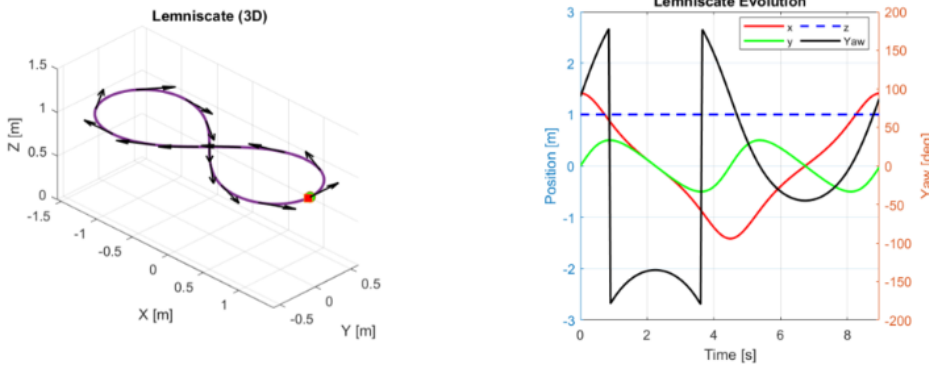


Figure 1. Visualization of the lemniscate trajectory:

(a) Lemniscate Trajectory in 3D space. (b) Evolution of x , y , z , and Ψ during $T_{\text{lemniscate}}$.

SUBSECTION 2.2

Spiral Ellipse Trajectory

The spiral ellipse trajectory describes a 3D elliptical helix motion, combining an elliptical path in the horizontal plane with a constant vertical ascent. This shape is highly effective for testing the controller's ability to handle varying curvature radius simultaneously with altitude changes. The trajectory is defined in `ShapeSpiralEllipse.m`.

The flat outputs for the position $\mathbf{z}(t)$ are defined as:

$$\begin{cases} z_1(t) = R_x \cos(\omega t) \\ z_2(t) = R_y \sin(\omega t) \\ z_3(t) = z_{\text{start}} + v_z t \\ z_4(t) = \text{atan2}(\dot{z}_2(t), \dot{z}_1(t)) \end{cases} \quad (2.4)$$

where R_x and R_y regulates the size of the ellipse, z_{start} is the initial altitude, v_z is the constant vertical velocity, and the yaw angle $z_4(t)$ is constrained to remain tangent to the horizontal path.

The first derivatives (velocities) are calculated as:

$$\begin{cases} \dot{z}_1(t) = -R_x \omega \sin(\omega t) \\ \dot{z}_2(t) = R_y \omega \cos(\omega t) \\ \dot{z}_3(t) = v_z \\ \dot{z}_4(t) = \frac{R_x R_y \omega^3}{\dot{z}_1(t)^2 + \dot{z}_2(t)^2} \end{cases} \quad (2.5)$$

The second derivatives (accelerations) are analytically derived as:

$$\begin{cases} \ddot{z}_1(t) = -R_x \omega^2 \cos(\omega t) \\ \ddot{z}_2(t) = -R_y \omega^2 \sin(\omega t) \\ \ddot{z}_3(t) = 0 \\ \ddot{z}_4(t) = -\frac{2R_x R_y \omega^3 (\dot{z}_1(t)\ddot{z}_1(t) + \dot{z}_2(t)\ddot{z}_2(t))}{(\dot{z}_1(t)^2 + \dot{z}_2(t)^2)^2} \end{cases} \quad (2.6)$$

The default configuration parameters for the simulated scenarios are:

- x -axis radius $R_x = 3.0$ m;
- y -axis radius $R_y = 2.0$ m;
- angular velocity $\omega = 0.7$ rad/s;
- initial altitude $z_{\text{start}} = 0.5$ m;
- vertical velocity $v_z = 0.2$ m/s;
- trajectory duration $T_{\text{ellipse}} = \frac{4\pi}{\omega}$ s, which corresponds to exactly two full elliptical revolutions while climbing.

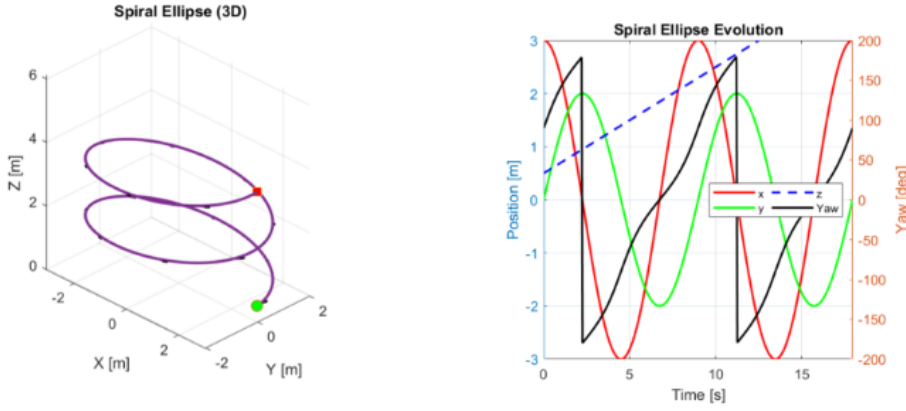


Figure 2. Visualization of the spiral ellipse trajectory:

(a) Spiral ellipse Trajectory in 3D space. (b) Evolution of x , y , z , and Ψ during $T_{SpiralEllipse}$.

SECTION 3

Open Loop Dynamics

To evaluate the impact of control implementation on the helicopter's spatial behavior, an open-loop simulation was made using the full nonlinear dynamic model. The primary purpose of this test is to compare the ideal trajectory, generated by a continuously updated control input $\mathbf{u}(t)$, against a trajectory driven by a Piece-Wise Constant (PWC) input $\mathbf{u}_k$. In the continuous case, the control actions are instantaneously derived from the differential flatness map. On the other hand, in the PWC approach the input is sampled at discrete steps and held strictly constant over the sampling interval $[kT_s, (k+1)T_s]$. The sequential state evolution is then computed by applying this discrete input to the nonlinear dynamic model at each time step.

The comparative analysis was performed across four complex flight scenarios (Circle, Lemniscate, Spiral, and Spiral Ellipse) evaluating multiple sampling times ($T_s \in \{0.02, 0.20, 0.50, 0.75\}$ seconds). The simulation has been performed within `OpenLooptest.m` script.

SUBSECTION 3.1

Circle Trajectory: Continuous vs Piece-Wise Constant Inputs

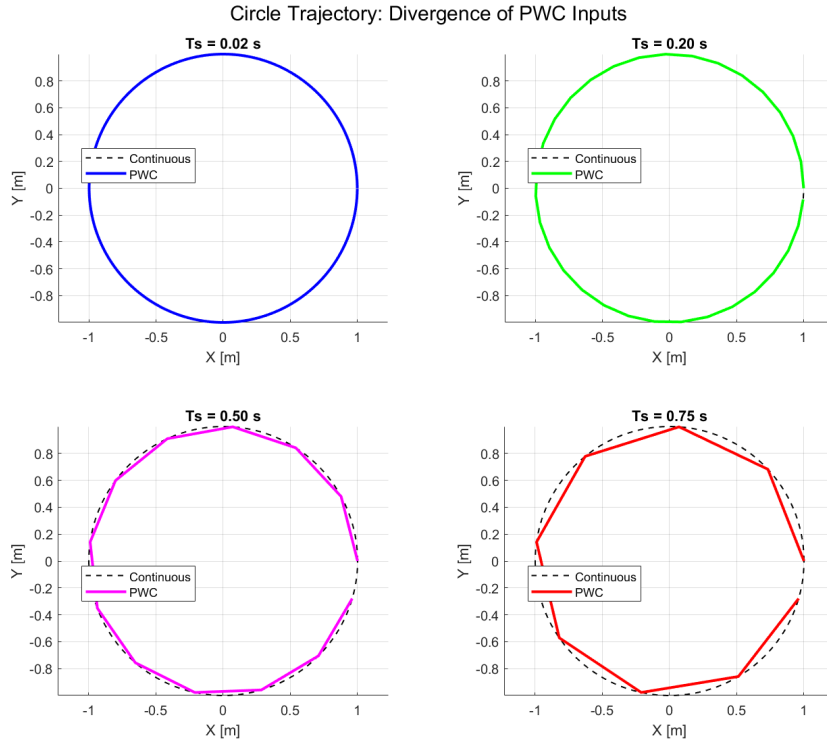


Figure 3. Circle Trajectory: Continuous vs Piece-Wise Constant Inputs reference trajectories over different values of T_s .

As illustrated in the figure above, increasing the sampling time T_s causes the Piece-Wise Constant (PWC) trajectory to progressively degrade and deviate from the ideal continuous circular path. Specifically, at larger intervals such as $T_s = 0.50$ s and $T_s = 0.75$ s, holding the control input constant over extended periods introduces significant errors, reducing the smooth curve to a distorted, polygonal approximation.

SUBSECTION 3.2

Lemniscate Trajectory: Continuous vs Piece-Wise Constant Inputs

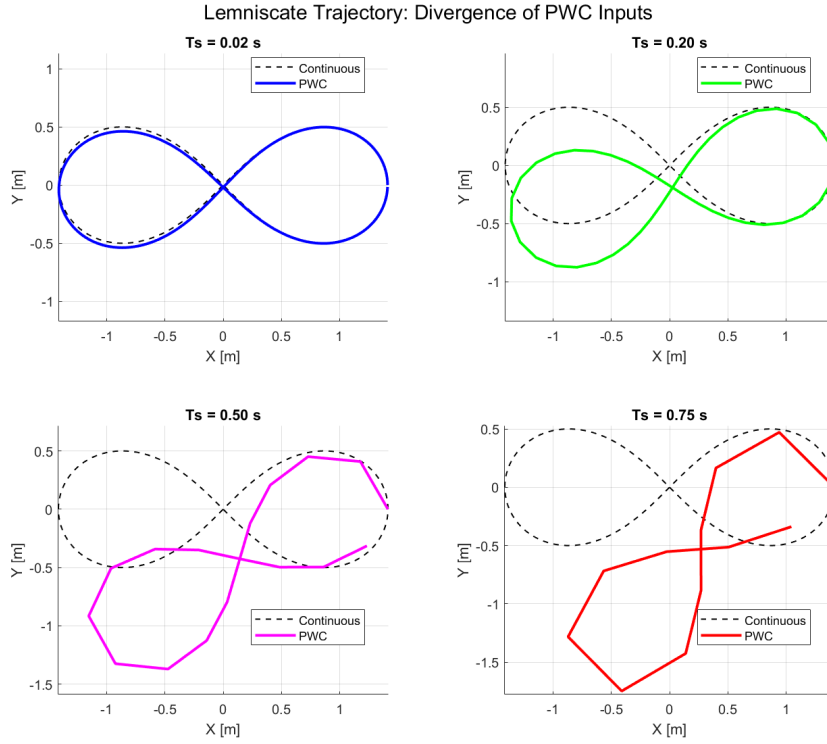


Figure 4. Lemniscate Trajectory: Continuous vs Piece-Wise Constant Inputs reference trajectories over different values of T_s .

The distortion of the Piece-Wise Constant (PWC) reference trajectory is caused by keeping the control inputs constant during each sampling interval T_s .

The lemniscate is a dynamic trajectory that requires continuous variations in heading and acceleration. When the nonlinear model is driven by inputs that are held constant for a longer time (e.g., $T_s = 0.50$ s or 0.75 s), it cannot smoothly follow these continuous changes.

As a result, the state evolution deviates from the ideal continuous-input solution. This deviation accumulates over time, progressively transforming the smooth figure-eight trajectory into a larger, asymmetric, and polygon-like shape.

SUBSECTION 3.3

Spiral Trajectory: Continuous vs Piece-Wise Constant Inputs

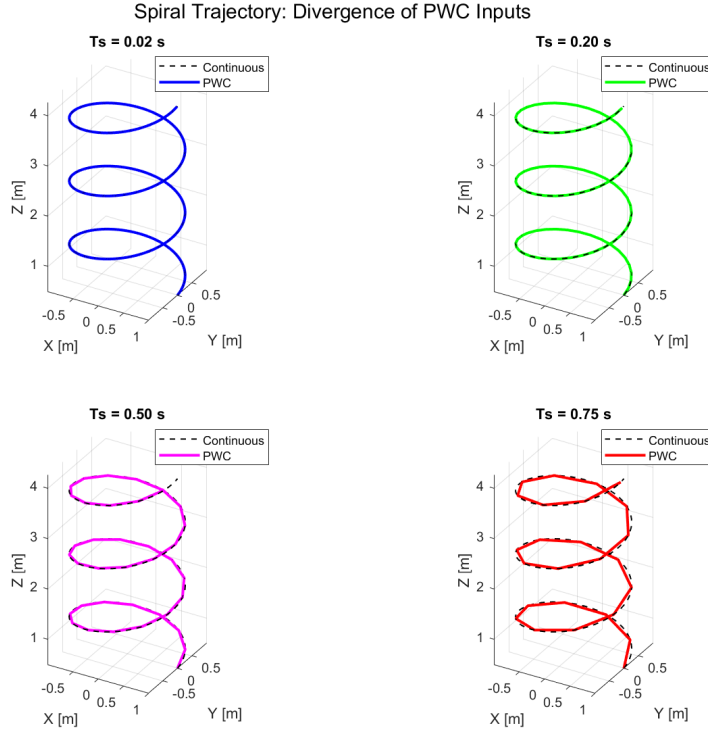


Figure 5. Spiral Trajectory: Continuous vs Piece-Wise Constant Inputs reference trajectories over different values of T_s .

In the spiral case, applying Piece-Wise Constant (PWC) inputs to the nonlinear model mainly produces a segmented approximation of the ideal continuous trajectory, rather than a strong geometric distortion as observed in the lemniscate case.

The spiral evolves more smoothly and does not require rapid changes in curvature. When the inputs are held constant over each sampling interval T_s , the model follows short motion segments that approximate the continuous evolution.

As T_s increases, the trajectory becomes visibly more polygonal and less smooth, but its overall shape and growth trend are preserved. Therefore, the effect of PWC inputs in this case is primarily a loss of smoothness, not a structural deformation of the path.

SUBSECTION 3.4

Spiral Ellipse Trajectory: Continuous vs Piece-Wise Constant Inputs

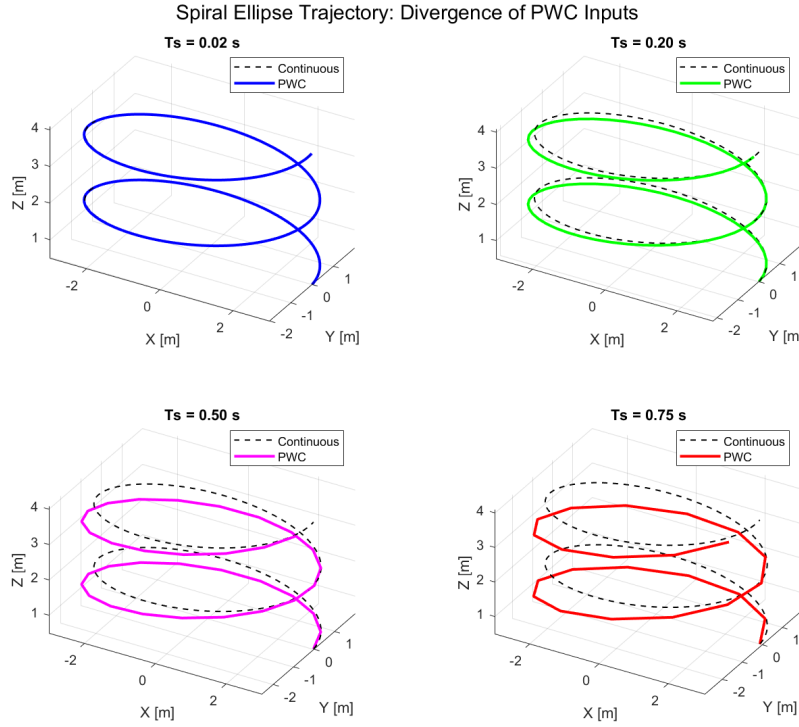


Figure 6. Spiral Ellipse Trajectory: Continuous vs Piece-Wise Constant Inputs reference trajectories over different values of T_s .

In the spiral ellipse case, the use of Piece-Wise Constant (PWC) inputs introduces both segmentation and geometric distortion as the sampling time T_s increases.

This trajectory combines vertical motion with non-uniform curvature due to the elliptical shape. When the nonlinear model is driven by inputs that are kept constant over each interval T_s , it cannot continuously adapt to the varying curvature.

For small T_s , the PWC trajectory closely matches the continuous solution. However, for larger T_s (e.g., 0.50 s or 0.75 s), the accumulated integration error becomes significant. The ellipse progressively deforms, and the spiral loses symmetry, resulting in a visibly distorted and polygonal path.

MPC Controller Modification

The modified controller is implemented in `MPCController_PWC_reference.m`. It is derived from the original `MPCController.m`, but it is adapted to work with a pre-generated Piece-Wise Constant (PWC) reference trajectory.

The distinction between the two implementations lies in the population of the nominal trajectory. In the original framework, the reference is computed online through the flatness map. In contrast, in the PWC implementation, the initialization method is the only component to be modified: the controller accepts a reference pair $(x_{\text{ref}}, u_{\text{ref}})$ produced offline by `GeneratePWC_reference.m` function.

The core LTV-MPC framework remains unchanged. The sequential linearization, discretization, QP formulation, cost structure, and warm-start strategy are identical in both implementations. The modification only affects how the reference trajectory is provided, enabling consistent tracking of piece-wise constant reference inputs.

PART

III

Experimental Results

Experimental Setup

The experimental results are presented in this section. All four reference trajectories are tested under different initial conditions and different sampling times ($T_s = 0.02, 0.05, 0.10, 0.20$ s).

The goal of using different initial conditions is to evaluate the robustness of the MPC controller and to challenge both the closed-loop optimization process and the tracking performance. Each scenario requires a specific initialization method.

Specifically, three initial conditions are considered:

On Reference In this scenario, the helicopter is initialized exactly at the first point of the reference trajectory. The objective of the MPC controller is to maintain accurate tracking for the entire simulation time. Since the system starts on the reference, the first LTV linearization is performed around the first reference state and the corresponding reference input, $[x_{\text{ref}}, u_{\text{ref}}]$. This represents the nominal operating point for the first optimization step.

Perturbation In this case, the helicopter is initialized with a deviation of 0.15 m on the x -axis with respect to the first point of the reference trajectory. The purpose of this test is to evaluate how the closed-loop system converges back to the desired trajectory for different sampling times.

The first LTV linearization is performed around the perturbed initial state and input obtained from differential flatness. The controller must therefore compensate for the initial tracking error and drive the system toward the reference trajectory.

Origin In the third scenario, the helicopter is initialized at the origin, with zero angles and zero velocities. This represents a more demanding condition, as the system is far from the reference trajectory and requires significant control action to reach it.

In this case, the first LTV linearization is performed around the zero state vector, which is used as the nominal operating point for the initial linearization. The controller

must first intercept the reference and then maintain trajectory tracking once convergence is achieved.

Remark For the **lemniscate trajectory** with the origin scenario, a different starting point has been selected. Since the lemniscate is centered at $(0, 0)$, initializing the system exactly at the origin would place it at the intersection of the two loops.

To obtain a more meaningful tracking test, the simulation is initialized at $x = 0.75$, which is approximately located at the center of the right loop of the lemniscate. The first LTV linearization is performed around this initial state. This implementation choice ensures that the tracking task starts from a representative condition rather than from the geometric center of the trajectory.

The default parameters used for this suite of experiments are faithful to the implementation in the original paper and include the following:

$$Q = Q_f = \text{diag}([50, 50, 50, 10, 3, 3, 1, 2, 15, 15]) \quad (5.1)$$

$$R = \text{diag}([2.0, 2.0, 2.0, 2.0]) \quad (5.2)$$

$$p = 18 \quad (5.3)$$

Remark Terminal cost was not employed for this simulation, setting $Q = Q_f$ as the authors suggest in the original paper. Model Simplification is not used, in order to analyze the complete behavior of the system.

The simulations were conducted using the script `MPCSimulationLoopPWC_reference.m`, specifically designed to evaluate the controller's robustness. The analysis focuses on the tracking quality of the trajectories while varying the sampling time $T_s \in \{0.02, 0.05, 0.10, 0.20\}$ and the initial conditions.

SECTION 6

Tracking Results

In the following sub sections, the results are presented separately for each reference trajectory. For each of them, a composite figure consisting of four subplots is provided. Each subplot corresponds to a different sampling time T_s , and within each subplot the three initial conditions are displayed simultaneously for comparison.

For clarity of presentation, only the final tracking trajectory snapshot is shown. The complete time evolution of each simulation has been recorded and exported in GIF format, available in the corresponding results directory.

Additionally, for each trajectory, a summary table reporting the quantitative performance metrics is provided. The reported metrics include State Root Mean Square Error (RMSE), State Mean Square Error (MSE), Input MSE, and the average solver computation time.

SUBSECTION 6.1

Circle Trajectory Results

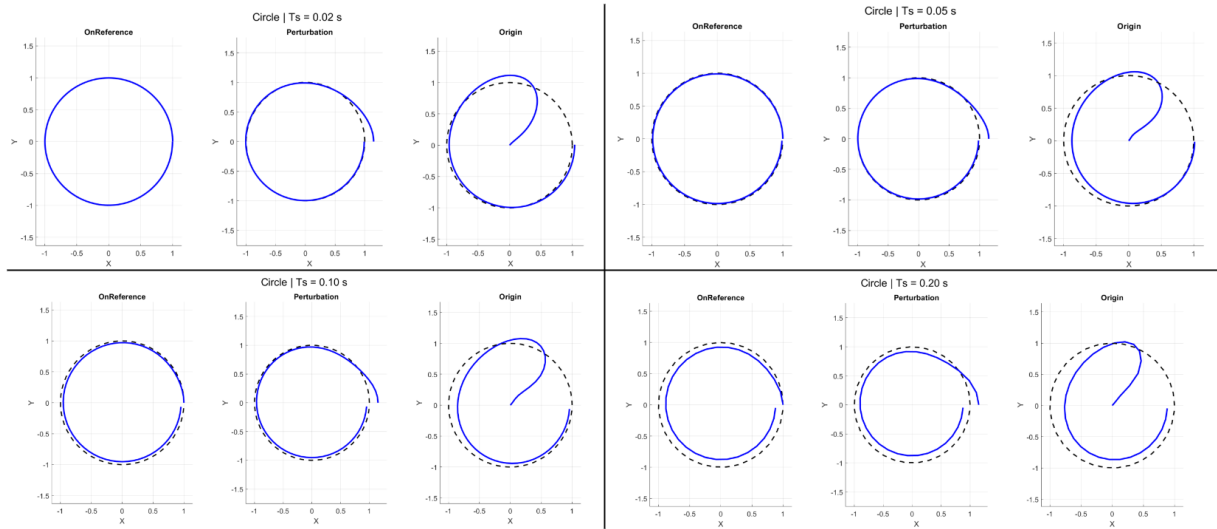


Figure 7. Circle Trajectory simulation.

Table 1. Quantitative Performance Metrics: Circle Trajectory for Different Sampling Times

T_s [s]	Initial Condition	State MSE	State RMSE	Input MSE	Avg. Solve Time [ms]
0.02	OnReference	6.54×10^{-6}	0.00256	3.43×10^{-6}	3.845
0.02	Perturbation	1.74×10^{-3}	0.04176	4.25×10^{-3}	3.813
0.02	Origin	3.19×10^{-1}	0.56503	1.53×10^{-1}	4.063
0.05	OnReference	1.68×10^{-4}	0.01295	1.58×10^{-4}	4.301
0.05	Perturbation	1.59×10^{-3}	0.03986	1.14×10^{-2}	4.180
0.05	Origin	2.89×10^{-1}	0.53728	2.40×10^{-1}	4.654
0.10	OnReference	1.86×10^{-3}	0.04318	2.57×10^{-3}	4.059
0.10	Perturbation	3.49×10^{-3}	0.05911	2.06×10^{-2}	4.470
0.10	Origin	3.26×10^{-1}	0.57129	2.75×10^{-1}	4.895
0.20	OnReference	1.19×10^{-2}	0.10890	1.90×10^{-2}	5.416
0.20	Perturbation	1.33×10^{-2}	0.11536	3.93×10^{-2}	4.643
0.20	Origin	4.22×10^{-1}	0.64954	1.998×10^{-1}	5.287

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SUBSECTION 6.2

Lemniscate Trajectory Results

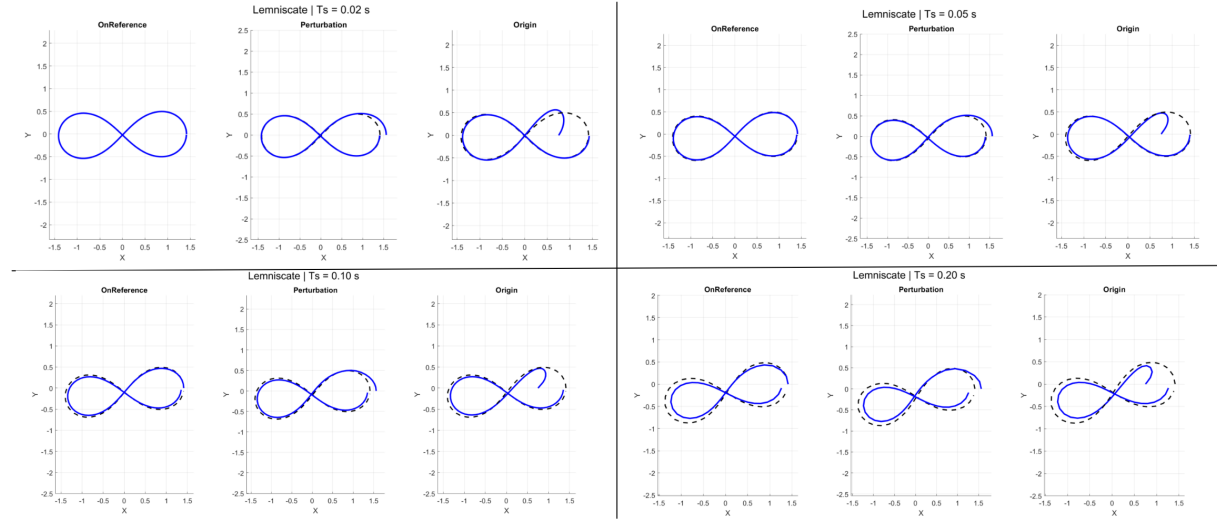


Figure 8. Lemniscate Trajectory simulation.

Table 2. Quantitative Performance Metrics: Lemniscate Trajectory for Different Sampling Times

T_s [s]	Initial Condition	State MSE	State RMSE	Input MSE	Avg. Solve Time[ms]
0.02	OnReference	7.82×10^{-6}	0.00280	1.84×10^{-5}	4.405
0.02	Perturbation	1.94×10^{-3}	0.04407	2.37×10^{-3}	4.373
0.02	Origin	1.74×10^{-1}	0.41753	6.04×10^{-2}	4.181
0.05	OnReference	2.02×10^{-4}	0.01422	5.31×10^{-4}	4.566
0.05	Perturbation	2.03×10^{-3}	0.04503	8.30×10^{-3}	4.709
0.05	Origin	1.54×10^{-1}	0.39279	8.15×10^{-2}	4.679
0.10	OnReference	1.75×10^{-3}	0.04185	4.71×10^{-3}	5.370
0.10	Perturbation	3.69×10^{-3}	0.06071	2.70×10^{-2}	5.081
0.10	Origin	1.82×10^{-1}	0.42623	9.44×10^{-2}	5.350
0.20	OnReference	8.31×10^{-3}	0.09113	2.96×10^{-2}	6.582
0.20	Perturbation	1.04×10^{-2}	0.10177	5.45×10^{-2}	6.247
0.20	Origin	2.62×10^{-1}	0.51173	6.06×10^{-2}	5.951

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SUBSECTION 6.3

Spiral Trajectory Results

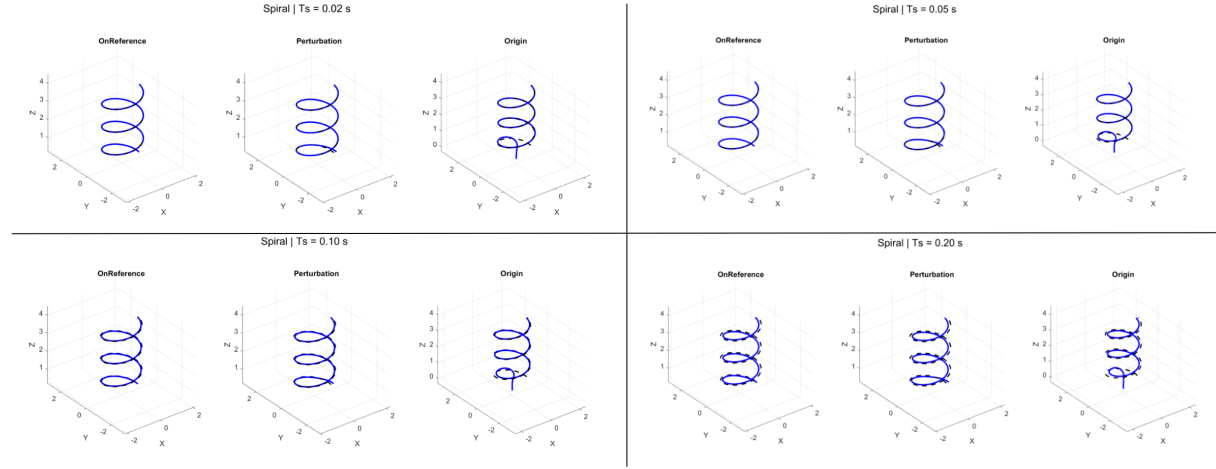


Figure 9. Spiral Trajectory simulation.

Table 3. Quantitative Performance Metrics: Spiral Trajectory for Different Sampling Times

T_s [s]	Initial Condition	State MSE	State RMSE	Input MSE	Avg. Solve Time [ms]
0.02	OnReference	7.37×10^{-6}	0.00272	2.57×10^{-6}	3.862
0.02	Perturbation	5.94×10^{-4}	0.02438	1.42×10^{-3}	3.732
0.02	Origin	9.46×10^{-2}	0.30756	5.02×10^{-2}	3.916
0.05	OnReference	1.93×10^{-4}	0.01390	9.13×10^{-5}	3.919
0.05	Perturbation	6.71×10^{-4}	0.02591	3.85×10^{-3}	3.943
0.05	Origin	8.36×10^{-2}	0.28922	7.92×10^{-2}	3.981
0.10	OnReference	2.21×10^{-3}	0.04700	1.30×10^{-3}	4.034
0.10	Perturbation	2.75×10^{-3}	0.05248	7.33×10^{-3}	4.088
0.10	Origin	9.62×10^{-2}	0.31020	9.10×10^{-2}	4.164
0.20	OnReference	1.42×10^{-2}	0.11913	9.25×10^{-3}	4.597
0.20	Perturbation	1.47×10^{-2}	0.12118	1.61×10^{-2}	4.258
0.20	Origin	1.35×10^{-1}	0.36805	6.98×10^{-2}	4.402

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SUBSECTION 6.4

Spiral Ellipse Trajectory Results

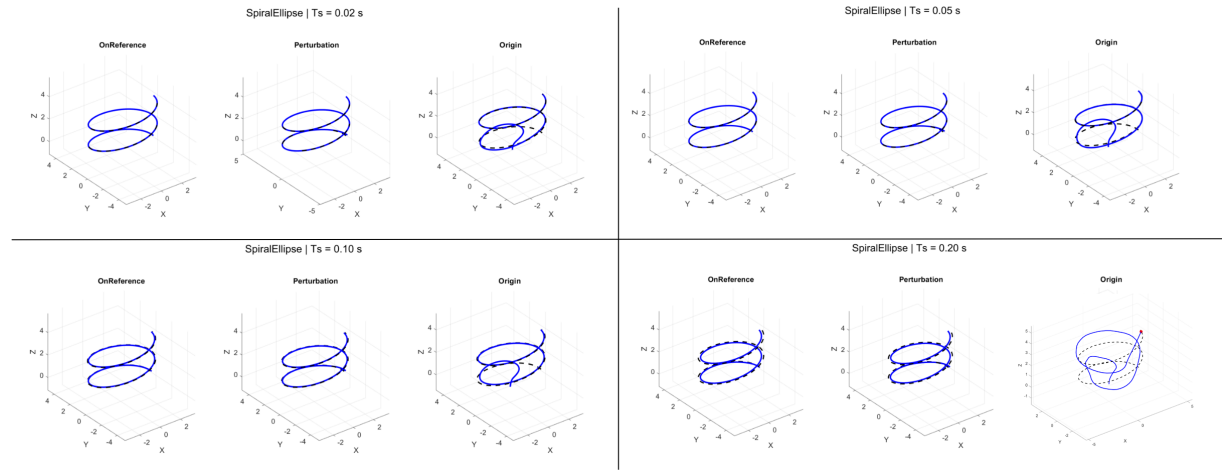


Figure 10. Spiral Ellipse Trajectory simulation.

Table 4. Quantitative Performance Metrics: SpiralEllipse Trajectory for Different Sampling Times

T_s [s]	Initial Condition	State MSE	State RMSE	Input MSE	Avg. Solve Time [ms]
0.02	OnReference	1.12×10^{-5}	0.00335	1.97×10^{-6}	3.891
0.02	Perturbation	6.66×10^{-4}	0.02581	1.55×10^{-3}	3.948
0.02	Origin	8.05×10^{-1}	0.89707	2.49×10^{-1}	4.457
0.05	OnReference	3.26×10^{-4}	0.01806	1.06×10^{-4}	4.241
0.05	Perturbation	9.65×10^{-4}	0.03106	3.10×10^{-3}	4.276
0.05	Origin	8.11×10^{-1}	0.90039	2.90×10^{-1}	4.647
0.10	OnReference	4.01×10^{-3}	0.06335	1.86×10^{-3}	4.237
0.10	Perturbation	5.00×10^{-3}	0.07074	7.16×10^{-3}	4.319
0.10	Origin	7.07×10^{-1}	0.84067	2.40×10^{-1}	4.689
0.20	OnReference	2.87×10^{-2}	0.16951	1.09×10^{-2}	4.550
0.20	Perturbation	3.04×10^{-2}	0.17424	1.91×10^{-2}	4.570
0.20	Origin	5.85	2.41826	1.95	7.887

SECTION 7

Overall Comparison

The comprehensive evaluation across the four proposed flight paths, Circle, *Lemniscate*, *Spiral*, and *Spiral Ellipse*, highlights the relationship between trajectory complexity, sampling time, and initial conditions when employing a Piece-Wise Constant (PWC) reference.

By generating the **PWC reference trajectory**, the controller is forced to track a path that inherently approximates the ideal continuous dynamics. For relatively smooth shapes like the circle and the spiral, the PWC approximation results mainly in a segmented, polygonal path. However, for highly dynamic shapes such as the lemniscate, which requires continuous variations in heading and acceleration to execute alternating coordinated turns, the constant actuation over each sampling interval introduces significant structural deviation.

The **initial conditions** play a key role in the overall tracking performance. When initialized on the reference, the MPC shows robust and stable tracking across all shapes, acting primarily to maintain the flight path on the reference trajectory. Introducing a spatial **perturbation** slightly increases the control effort and the resulting state errors, but the system consistently proves capable of recovering the desired path quickly. In contrast, initializing the helicopter at the **origin** represents the most challenging scenario. The system starts far from the reference, requiring aggressive maneuvering and significant control action to intercept the moving reference.

A general trend observed in the data is that increasing the **sampling time** T_s degrades the tracking accuracy, as longer intervals hold the inputs constant for too long, increasing the geometric mismatch with the ideal path.

Interestingly, for specific scenarios such as the Circle (Perturbation) and Lemniscate (Origin), the tables show that increasing the sampling time from $T_s = 0.02$ s to 0.05 s actually reduces the state RMSE. This suggests that the tracking performance is highly sensitive to the magnitude of the PWC control action. In these cases, a larger T_s facilitates a more robust correction by applying a higher PWC input over the sampling period. This allows the non-linear system to react to the initial errors more aggressively, whereas the lower inputs associated with a smaller T_s result in a slower and less efficient approach to the desired path.

The most critical scenario observed in the experiments is the Spiral Ellipse at $T_s = 0.20$ s initialized at the origin, which results in the highest recorded error. This severe drop in tracking performance happens because the higher sampling time is insufficient to accurately capture and follow the varying curvature of such a complex 3D shape, especially when trying to recover from a distant starting point.

The control algorithm remains highly computationally efficient across all tests. The **average solving time** consistently stays between 3 and 5 ms, with only a minor spike up to nearly 8 ms. Moreover, an analysis of the **Input MSE** reveals that while severe initial conditions (such as starting from the origin) demand a much higher and more aggressive control effort to reach the reference, the solver handles these large control variations without noticeable delays. This proves that, despite the challenges introduced by large initial tracking errors, the underlying QP formulation and warm-start strategy remain robust, making the controller suitable for real-time, high-frequency applications.

SECTION 8

Convergence Study

To further evaluate the tracking performance of the MPC controller, this section analyzes the **position error (m)** for each reference trajectory during the simulation. For each trajectory shown in the following plots, four different sampling times are considered:

$$T_s \in \{0.02, 0.05, 0.10, 0.20\} \text{ s.}$$

The objective of this analysis is to provide a quantitative assessment of the convergence behavior in each scenario, with a clearer evaluation of the tracking performance.

For each simulation step, with the total number of steps defined as

$$N = \frac{T_{\text{simulation}}}{T_s},$$

the position error is computed as the Euclidean distance between the simulated position and the reference position:

$$e_{\text{pos}} = \sqrt{(x_{\text{sim}} - x_{\text{ref}})^2 + (y_{\text{sim}} - y_{\text{ref}})^2 + (z_{\text{sim}} - z_{\text{ref}})^2}.$$

In addition, the thresholds $\{0.05, 0.10, 0.20, 0.50\}$ are included in the plots to visually verify the convergence of the tracking error. For each trajectory, the three previously discussed initial conditions are analyzed.

SUBSECTION 8.1

Circle Trajectory

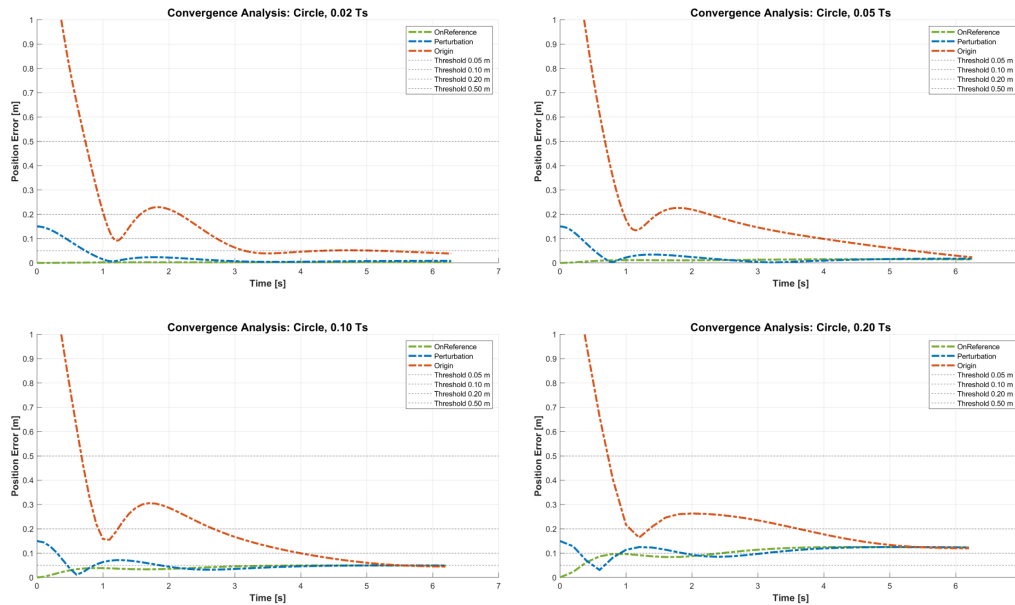


Figure 11. Evolution of Position Error during the simulation, circle trajectory.

SUBSECTION 8.2

Lemniscate Trajectory

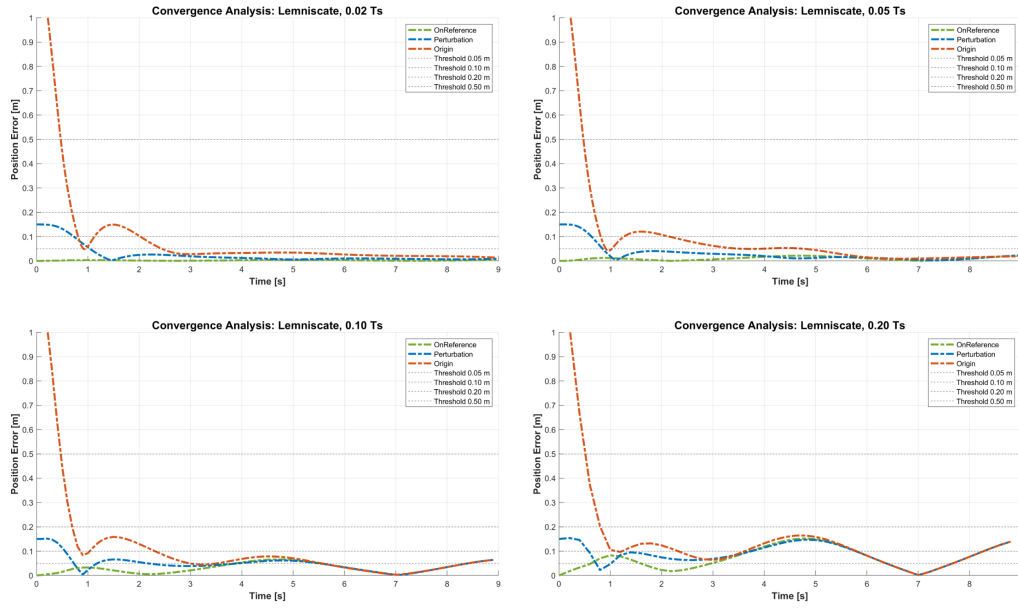


Figure 12. Evolution of Position Error during the simulation, lemniscate trajectory.

SUBSECTION 8.3

Spiral Trajectory

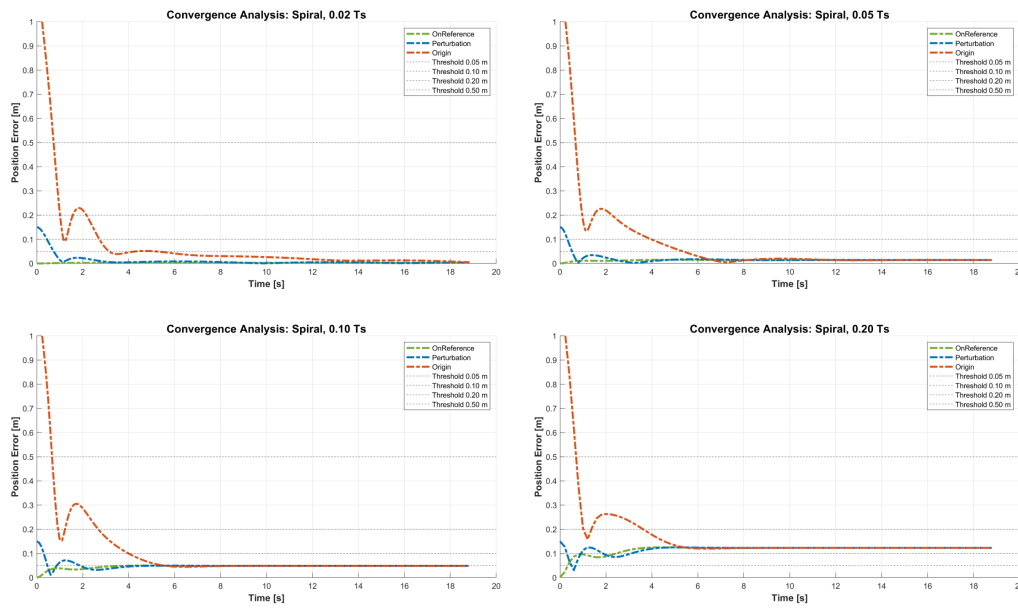


Figure 13. Evolution of Position Error during the simulation, spiral trajectory.

Spiral Ellipse Trajectory

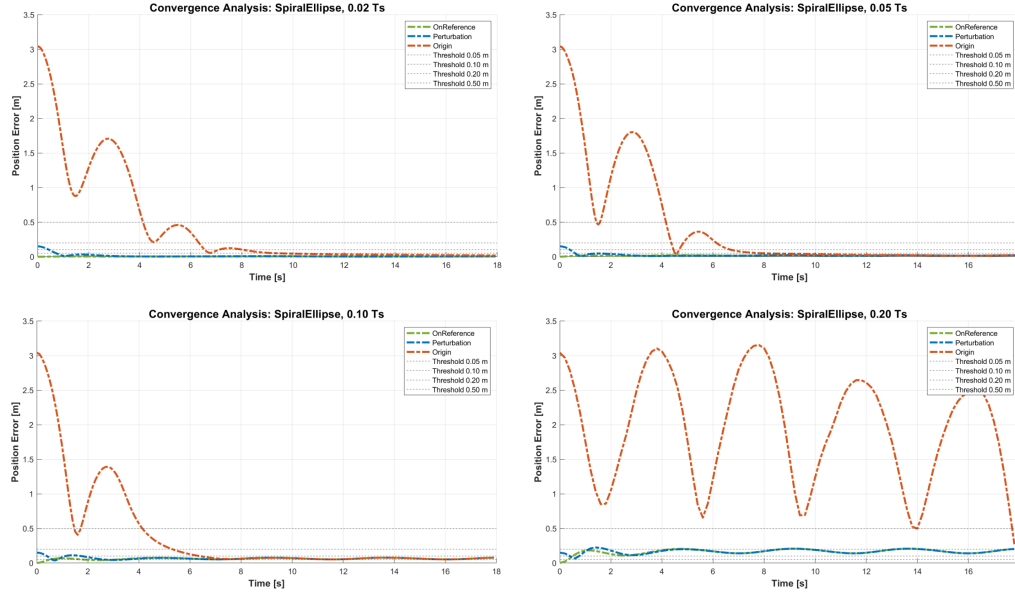


Figure 14. Evolution of Position Error during the simulation, spiral ellipse trajectory.

Results of the Convergence Study

For all trajectories, when the system is initialized *On Reference*, the position error remains very small for every sampling time. In this case, the controller aims to keep the helicopter on the reference. As expected, increasing the sampling time T_s results in a slower convergence and a higher steady-state error.

Under the *Perturbation* condition, the controller successfully drives the system back to the desired trajectory in all scenarios. For small sampling times ($T_s = 0.02$ s and 0.05 s), the convergence is generally faster and smoother. For larger sampling times ($T_s = 0.10$ s and 0.20 s), convergence is still achieved, but with a slower transient response and a larger error.

The *Origin* condition represents the most challenging case. For the Circle and Spiral trajectories, the controller is able to follow the reference and reduce the position error below the thresholds within a reasonable time for all sampling times. The Lemniscate trajectory shows a slightly more oscillatory convergence, due to its alternating curvature and continuous heading variations. The most critical behavior is observed for the Spiral Ellipse with $T_s = 0.20$ s, where the combination of complex 3D motion and large initial distance significantly degrades the convergence performance, leading to the highest error values among all experiments.

Overall, the convergence study confirms that the MPC is robust with respect to different initial conditions and trajectory geometries. Smaller sampling times generally improve accuracy and speed of convergence, while larger sampling times reduce tracking precision, especially for complex trajectories. The results show that the controller achieves convergence of the position error in most of the tested configurations, while

performance degradation appears only in the most demanding cases, characterized by complex motion and large sampling times.

Conclusions

SECTION 9

Conclusions

This project extended the original *MPC framework for miniature helicopters* for trajectory tracking by introducing additional reference trajectories and adopting a Piece-Wise Constant (PWC) formulation for reference generation.

In particular, new complex flight paths, such as the *Lemniscate* and the *Spiral Ellipse*, were analytically derived using differential flatness, allowing the computation of dynamically consistent state and input references. An **open-loop analysis** was performed to compare continuous-time inputs with PWC, showing that increasing the sampling time produces geometric distortions and reduced smoothness, since the control inputs are kept constant over longer sampling intervals.

The MPC controller was modified to directly track pre-generated PWC reference trajectories, while preserving the original LTV-MPC structure, cost formulation, and QP solution. **Closed-loop simulations** were performed under different sampling times and initial conditions (On Reference, Perturbation, and Origin), providing an evaluation of robustness and tracking capability. Quantitative performance metrics, including State MSE, RMSE, Input MSE, and average solver time, confirmed that the controller maintains high computational efficiency while ensuring accurate tracking in most scenarios.

The results clearly show that both the sampling time and the initial condition significantly influence performance. Smaller sampling times generally improve tracking accuracy, whereas larger sampling times may degrade performance because the control input is updated less frequently. The **convergence study** further confirmed that the controller is able to reduce the position error below predefined thresholds in the majority of cases, with performance degradation mainly observed for complex trajectories combined with less frequent sampling. Overall, this extension demonstrates the robustness, flexibility, and real-time applicability of the proposed MPC framework when tracking PWC reference trajectories.